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DECLARATION OF AUTHENTICITY

Assignment:	KM01. Introduction to Data Science and Data Analysis. Learner Workbook
Date Submitted:	15 October 2025
Qualification / Skills Programme:	Occupational Certificate: Data Science Practitioner | SAQA ID: 118708
Assessment Site / Provider:	Praesignis PTY LTD | MICT SETA

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I, the above-mentioned learner, hereby declare that:

- The attached submission is my own original work.
- I have acknowledged all sources used in the completion of this work.
- I have not allowed, and will not allow, any other person to copy my work.
- ✓ I understand that false declarations constitute fraud and will be subject to disciplinary procedures and nullification of results by the QCTO.
- ✓ I understand that this assignment/portfolio is my own original work and I have not previously submitted it, in its entirety or in part, for obtaining any qualification.

Learner Signature: **Date:** 15 October 2025

Assignment Portfolio

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Assignment Details

Assignment: KM01. Introduction to Data Science and Data Analysis. Learner Workbook

Submission Date: Wednesday, 15 October 2025, 10:47 AM

Grade: 84.96 / 100.00 (84.96%)

Graded: Wednesday, 4 March 2026, 1:59 PM

Submitted Files

- KM-01 Learner workbook.pdf

Marking Guide Assessment

Final score: 96.00 / 113.00 (84.96%)

KM-01-KT01: What is Data Science?

Description: 1. What is business intelligence?

Markers: Business Intelligence (BI) is a technology-driven process for analyzing data and presenting actionable information to help organizations make informed decisions. (1)
It involves collecting, governing, and transforming data into insights using tools like dashboards and reports. (1)
BI focuses on descriptive analysis (what happened) and is used to improve processes, uncover revenue opportunities, and stay ahead of competitors. (1)

Score: **2.00 / 3.00 (66.67%)**

KM-01-KT01:

Description: 2. What Is Data Analysis?

Markers: Data analysis is the process of collecting, cleaning, modeling, and interpreting data to extract meaningful insights. (1) It involves:
Descriptive analysis: Summarizing historical data (e.g., sales reports). (1)
Predictive analysis: Forecasting trends (e.g., machine learning models). (1)
Prescriptive analysis: Recommending actions (e.g., optimizing supply chains). (1)
Example: Netflix analyzes viewing data to recommend shows. (1)

Score: **2.00 / 5.00 (40.00%)**

Remark: You were supposed to provide the different types of analysis, e.g., descriptive, predictive, and prescriptive.

KM-01-KT01:

Description: 3. Describe current global trends in Big Data and explain their impact.

Markers: Trends:
AI and Machine Learning: Automating data analysis (e.g., fraud detection in banking). (2)
IoT Growth: Sensors generate real-time data (e.g., smart meters in Cape Town's drought management). (2)
Cloud Computing: Enables scalable storage/processing (e.g., Microsoft's Project Brainwave). (2)
Impact:
Healthcare: Early disease detection (e.g., Google's DeepMind for cancer diagnosis). (2)
Retail: Personalized recommendations (e.g., Amazon's product suggestions). (2)
Public Services: Crime prediction using SAPS data. (2)

Score: **4.00 / 12.00 (33.33%)**

Remark: Not enough information was provided to get the full marks.

KM-01-KT02: Why data science?

Description: 1. Explain how data science helps the economy.

Markers: Informs policy: Economists use data to address societal issues (e.g., unemployment, inflation). (2)
Sector growth: Drives innovation in finance (fraud detection), agriculture (crop optimization), and logistics (route efficiency). (2)
Job creation: Demand for data professionals boosts employment. (2)

Score: **4.00 / 6.00 (66.67%)**

KM-01-KT02:

Description: 2. Describe how data science helps society and communities.

Markers: Public benefits: Helps vulnerable families access services faster (e.g., welfare programs). (2)
Crisis response: Predicts droughts (e.g., Cape Town's Day Zero) or disease outbreaks (e.g., Ebola). (2)
Nonprofits: Improves outreach and resource allocation (e.g., refugee resettlement). (2)

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT02:

Description: 3. Describe how data science helps businesses.

Markers: Predictability: Forecasts trends (e.g., sales, customer churn). (2)
Real-time intelligence: Dashboards for quick decisions (e.g., inventory management). (2)
Marketing: Personalizes customer experiences (e.g., Spotify's recommendations). (2)
Security: Detects fraud (e.g., PayPal's anomaly detection). (2)

Score: **8.00 / 8.00 (100.00%)**

KM-01-KT03: Sources of data

Description: 1. Explain the concept of data capture.

Markers: Data capture is the process of collecting and converting raw data (e.g., forms, sensor readings) into digital formats for analysis. (1) It can be:
Manual: Typing survey responses. (1)
Automated: Barcode scanners, IoT sensors. (1)

Score: **1.00 / 3.00 (33.33%)**

KM-01-KT03:

Description: 2. Define a data capture system.

Markers: A system that collects, records, and stores data from sources like: (1)
Barcode scanners: Retail checkout. (1)
Biometric sensors: Fingerprint login. (1)
APIs: Pulling weather data. (1)

Score: **3.00 / 4.00 (75.00%)**

KM-01-KT03:

Description: 3. List and describe different methods of capturing data.

Markers: Method: Example - Use Case
Manual Entry: Typing customer details - Call centers
OCR: Scanning ID documents - Bank onboarding
Biometrics: Facial recognition for SIM cards - Mobile networks
IoT Sensors: Smart meters - Utility monitoring

Score: **9.00 / 9.00 (100.00%)**

KM-01-KT04: Ensuring access to accurate data

Description: 1. What is electronic identity verification?

Markers: eIDV confirms a person's identity by matching their data (e.g., ID number, selfie) against trusted databases. (1)
Steps:
Submit ID document. (1)
AI checks authenticity. (1)
Liveness detection (e.g., blinking). (1)
POPIA-compliant storage. (1)
Example: Banks use eIDV for online account opening. (1)

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT04:

Description: 2. Why is identity verification software Important?

Markers: Prevents fraud: Stops fake accounts. (2)
Compliance: Meets POPIA/GDPR laws. (2)
Customer experience: Faster onboarding. (2)

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT04:

Description: 3. Explain document data capture with an example.

Markers: Process:
Scan a South African ID. (1)
OCR extracts name, ID number, DOB. (1)
Data is stored for verification. (1)
Example: Telecom companies use OCR to verify customers during SIM registration. (1)

Score: **3.00 / 3.00 (100.00%)**

KM-01-KT04:

Description: 4. Identify the type of data that should be indexed.

Markers: Index these for quick retrieval:
Unique IDs (e.g., employee numbers). (2)
Dates (e.g., transaction timestamps). (2)
Geographic data (e.g., customer provinces). (2)
Avoid indexing: Large text fields (e.g., comments). (or (2))

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT05: Applications of data science

Description: 1. Describe how data science supports decision-making in business.

Markers: At least 3 examples x 2 = 6 marks
Competitive edge: Analyzes customer behavior to stock trending products. (2)
Optimization: Reduces costs via supply chain analytics. (2)
Consumer experience: Recommends products (e.g., Netflix). (2)
Healthcare: Predicts patient risks for early intervention. (2)

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT06: Attributes of a Data Science Practitioner

Description: 1. Explain how the following skills elevate the quality and impact of data science.
• Oral and written communication skills
• Attention to detail
• Analytical thinking
Markers: Communication: Translates technical insights for stakeholders. (1)
Attention to detail: Ensures clean data and accurate models. (1)
Analytical thinking: Solves complex problems (e.g., clustering customer segments). (1)

Score: **3.00 / 3.00 (100.00%)**

KM-01-KT06:

Description: 2. Identify and explain the key attributes and attitudes required for a Data Science practitioner to progress in their career path.
Markers: Curiosity: Learns new tools (e.g., Python, TensorFlow). (2)
Business acumen: Aligns projects with organizational goals. (2)
Ethics: Ensures fair, unbiased algorithms. (2)

Score: **6.00 / 6.00 (100.00%)**

KM-01-KT07: Big Data

Description: 1. Describe the four vs of Big Data, i.e., volume, velocity, variety, and veracity.

Markers: V: Meaning - Example
Volume: Massive data (terabytes+) - Netflix’s 270M user profiles (3)
Velocity: High-speed data (real-time streams) - Stock market trades (3)
Variety: Structured/unstructured data - Social media posts + SQL tables (3)
Veracity: Data quality/trustworthiness - Cleaning sensor errors (3)

Score: **12.00 / 12.00 (100.00%)**

KM-01-KT07:

Description: 2. List and describe common sources of Big Data

Markers: Social Data: Likes, tweets (e.g., Twitter trends). (3)
Machine Data: IoT sensors (e.g., smart water meters). (3)
Transactional Data: Invoices, payments (e.g., retail sales). (3)

Score: **9.00 / 9.00 (100.00%)**

DATA SCIENCE PRACTITIONER - LEARNER WORKBOOK

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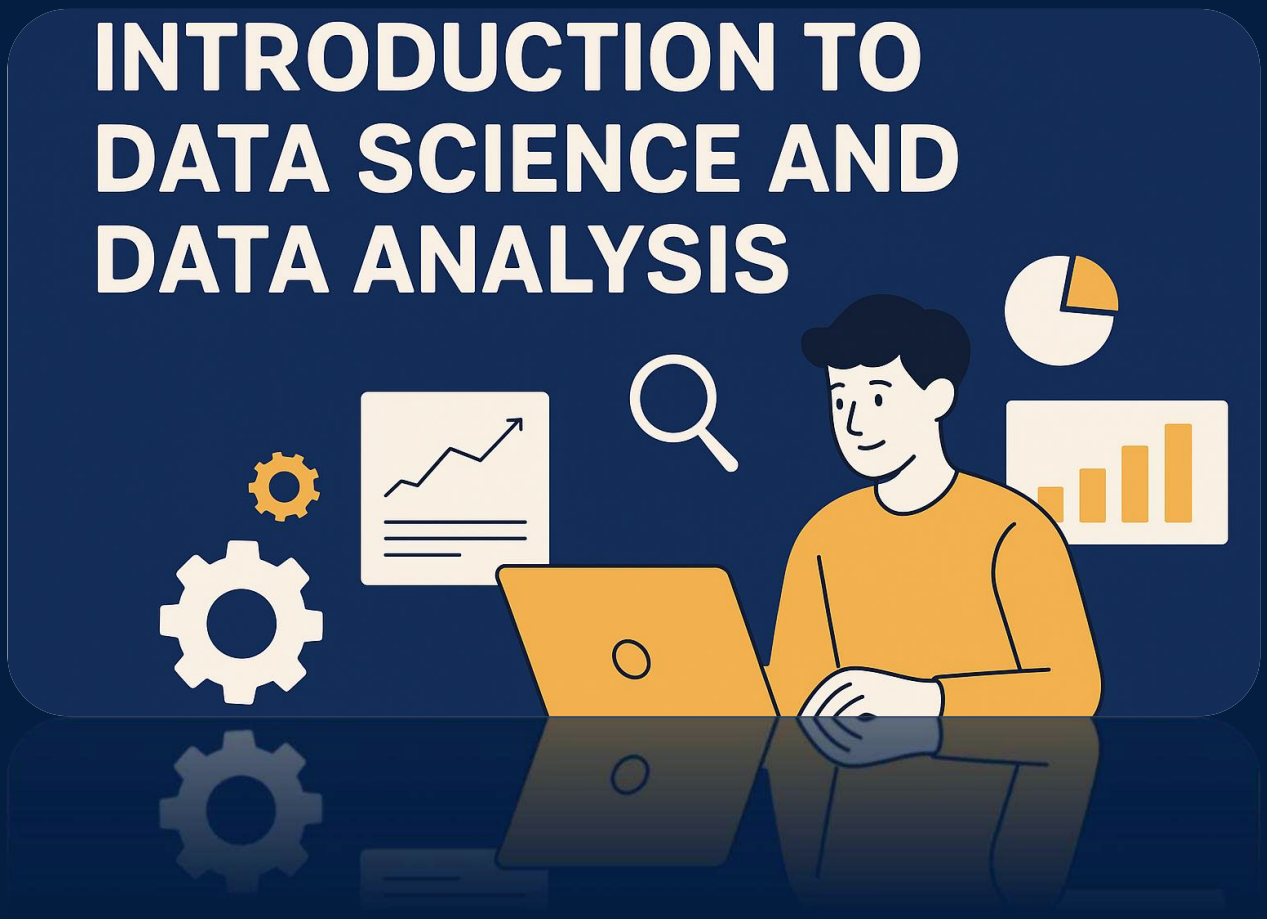
Connecting data, people, and the cloud.

KNOWLEDGE MODULE 01: INTRODUCTION TO DATA SCIENCE AND DATA ANALYSIS



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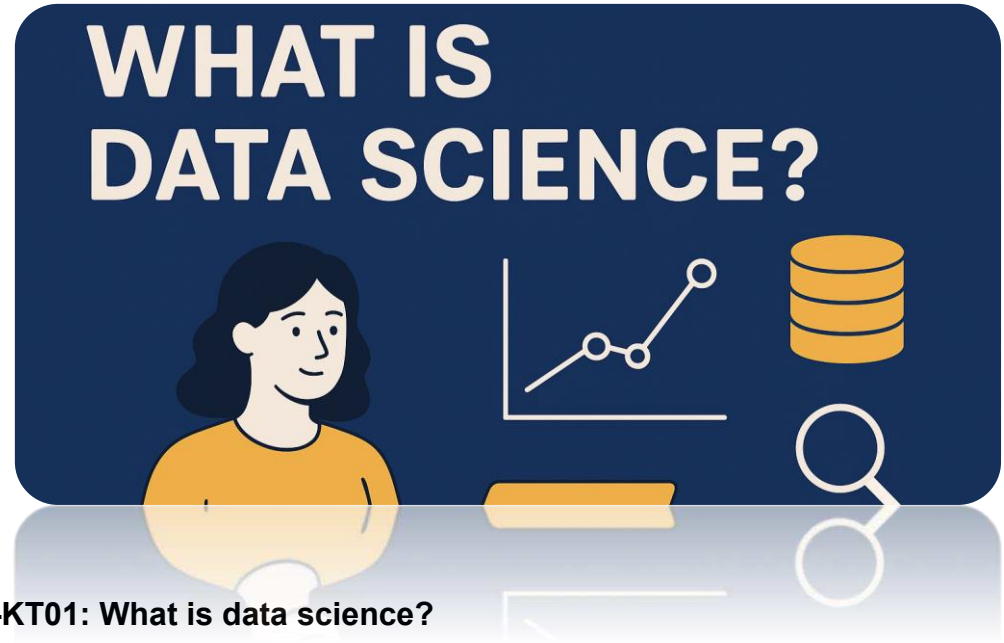

KM-01-KT01: What is data science?

1. **What is business intelligence? 3 marks**

Business Intelligence (BI) is a method of doing descriptive analysis of data with technology and skills to help make informed business decisions. It uses a set of tools for BI that collect, govern, and transfer data.

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2. **What Is Data Analysis? 5 marks**

Data analysis is the process of extracting insights and meaningful patterns from data through the use of statistical and analytical techniques, tools, and methods. It involves cleaning, transforming, and modelling data to discover relevant information, draw conclusions, and support informed decision-making.

3. **Describe current global trends in Big Data and explain their impact. 12 marks**

YouTube's videos are and example of Big data.

Big data trends used: AI and ML integration, data analytics, ad real-time processing.

YouTube collects numerous amounts of data on user behaviour such as:

- Watch history.
- Search queries.
- Likes and dislikes.
- Engagement metrics.
- Device and platform information.

Impact: Personalized video recommendations enhance user experience, increase engagement, and drive content consumption. This data driven approach has contributed to YouTube's massive user base and revenue growth.

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KM-01-KT02: Why data science?

1. **Explain how data science helps the economy. 6 marks**

Data science helps the economy by:

- Helping economist to solve societal issues both big and small.
- Allowing economists to apply theoretical concepts to practical situations and leverages large datasets to make informed decisions in sectors such as finance, government, entertainment, and industry.
- Tracking and managing resources efficiently, supporting green energy initiatives, reducing waste, and monitoring environmental changes, all of which contribute to long-term economic sustainability.

2. **Describe how data science helps society and communities. 6 marks**

Data science helps the society and communities by.

- Helping NGOs to achieve more with their data by improving their services, interventions, and outreach so they can fulfil their mission of bettering the world and people's lives.
- Accessing public benefits saving water and money during droughts, and saving time in resettling refugees, so they can find homes and jobs faster.
- Supporting societies and communities by enabling smarter decisions, fostering equality, and enhancing access to essential services.

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3. **Describe how data science helps businesses. 4 x 2 = 8 marks**

Data science helps businesses by:

- Helping businesses monitor, manage, and collect performance measures to improve decision-making across the organization.
- Using trend analysis to make critical decisions to improve consumer engagement, corporate performance, and boost revenue.
- Increasing business predictability, increased predictability allows businesses to make informed decisions that can lead to better outcomes.
- Ensuring real-time intelligence, this kind of intelligence is important for allowing managers to make faster and more informed decisions.

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KM-01-KT03: Sources of data

1. **Explain the concept of data capture. 3 marks**

Data capture refers to the process of collecting and extracting data from different sources such as, surveys, sensors, and apps. ✓

2. **Define a data capture system. 4 marks**

Data capture system is a software or technology solution that collects, extracts and processes data from different sources automating the data collection process. ✓

- Collect data from multiple sources. ✓
- Validate and verifies data for accuracy. ✓
- Formats and standardize data for use.

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3. **List and describe different methods of capturing data. 3 x 3 = 9 marks**

1. Barcode Scanning: Capturing data from barcodes using a scanner.
2. Sensor Data Capture: Collecting data from sensors and IoT devices.
3. Web Scrapping: Extracting data from websites and online platforms.

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ENSURING ACCESS TO ACCURATE DATA

KM-01-KT04: Ensuring access to accurate data.

1. **What is electronic identity verification? 6 marks**

Electronic identity verification is a digital process that verifies an individual's identity using electronic data and documents. For example, verifying identity documents like ID books and passports. Complies with regulatory requirements like FICA ad POPIA. Electronic identity verification prevents identity theft, money laundering, and other financial crimes, while ensuring compliance with regulations and protecting sensitive information.

2. **Why is identity verification software Important? 3 x 2 = 6 marks**

Prevents identity theft: Protects individuals and organizations from identity- related crimes.

Reduced fraud: Verifies identities to prevent fraudulent activities and transactions.

Endures compliance: Meets regulatory requirements (e.g., FICA, POPIA) and industry standards.

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3. **Explain document data capture with an example. 3 marks**

Document data capture is the process of extracting data from documents, such as invoices, contracts, or forms using automated technologies like optical character recognition. For example, scan or upload invoices, extract data, validate data for accuracy and export data to their accounting system.

4. **Identify the type of data that should be indexed. 3 x 2 marks**

1. Unique identifiers: For quick look up and data linking.
2. Date and timestamps: Used in time-based filters and reports.
3. Categorical fields: Useful for filtering and grouping.

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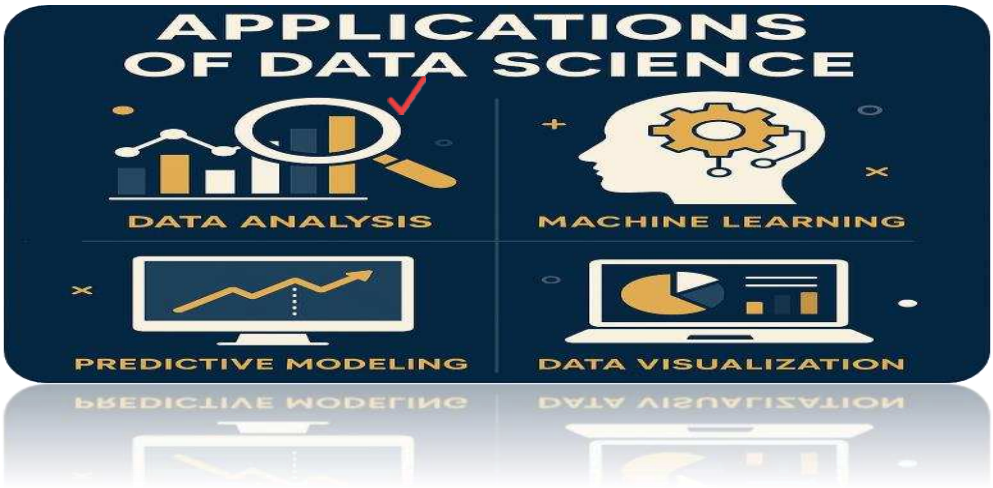

KM-01-KT05: Applications of data science

1. **Describe how data science supports decision-making in business. At least 3 examples x 2 = 6 marks**

Data science can improve decision making by understanding how choices affect outcomes.

Examples:

- For businesses data scientists are responsible for turning raw data into cooked data, they also help to analyse the health of the business.
- For customer experience it recommends products based on the customer preferences and improving satisfaction.
- For agriculture they use sensors and AI to improve to improve crop yields and reduce resource waste.

2.

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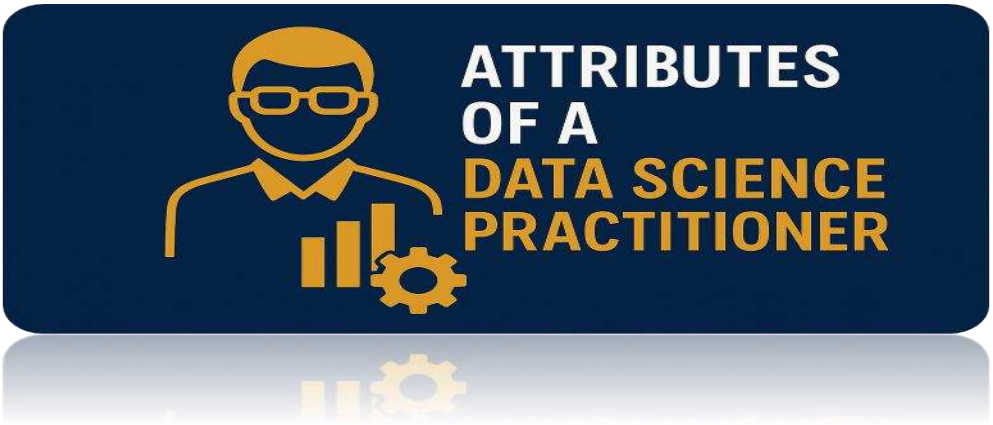

KM-01-KT06: Attributes of a Data Science Practitioner

1. **Explain how the following skills elevate the quality and impact of data science. 3 x 1 = 3 marks**

   - Oral and written communication skills
   - Attention to detail.
   - Analytical thinking

Oral and written communication skills: Effective communication bring it all together, translating technical work into real-work decisions and aligning data initiatives with business goals. ✓

Attention to detail: Ensures data is clean, models are accurate, and errors don't slip through unnoticed. ✓

Analytical thinking: A sharp analytical mind allows a data scientist to ask the right questions and choose the suitable models. ✓

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2. **Identify and explain the key attributes and attitudes required for a Data Science practitioner to progress in their career path. 3 x 2 = 6 marks**

Curiosity and Growth Mindset: Eagerness to learn, explore and stay updates with new tools and techniques.

Analytical Thinking: Ability to break down problems and interpret logically.

Technical Proficiency: Competence in programming, data analysis, machine learning and tools like python, SQL, etc.

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KM-01-KT07: Big Data

1. **Describe the four vs of Big Data, i.e., volume, velocity, variety, and veracity. 4 x 3 = 12 marks**

Volume: The amount of data generated, stored, and analysed.

Velocity: The different types and sources of data (Structured and Unstructured).

Variety: The speed at which data is created, shared, and processed.

Veracity: The trustworthiness and quality of data.

2. **List and describe common sources of Big Data. 3 x 3 = 9 marks**

Social data: It provide invaluable insights into consumer behaviour and sentiment that can be greatly influential in marketing analytics. For example, the public web and tools like Google trends can used to increase the volume of big data.

Machine data: Is information that is generated by industrial equipment. For example, sensors that are installed in machinery and web logs that track user behaviour. This type of data grows really well as the internet of things grows and expands around the world.

Transactional data: Is generated from daily transactions that takes place both online and offline. For example, invoices, payments orders, and storage records.

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REFERENCE

://www.upgrad.com › › Big Data

...https://www.forbes.com › sites

://www.techtarget.com › feature › 6-big-data-benefits...

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